Lab 03
Cascode Amplifier

Analog IC Design Course – Cadence Tools (Self Study)

Student Information

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Under Guidance and Help of My More Experienced Friends

Labs Provided by **Dr. Hesham Omran** via Master-Micro.com

Thanks to everyone who has helped me,
I pledge to help others as well.

Done during my **Post Graduate MSc. Program** in
Electronics and Electrical Communication Engineering
at the **Faculty of Engineering – Ain Shams University**

Lab 03: Cascode Amplifier

Intended Learning Objectives:[1]

In this lab[2] you will:

- Design and simulate a cascode amplifier.
- Learn how to generate and use design charts.
- Study the biasing of the cascode amplifier.
- Investigate the gain, the bandwidth, and the GBW of a cascode amplifier.

Part 1: Sizing Chart

1) We can show that the intrinsic gain of a MOSFET is given by: $|Av| \approx g_m r_o = \frac{2I_D}{V_{ov}} \times \frac{V_A}{I_D} = \frac{2V_A}{V_{ov}}$

Interestingly, the gain only depends on λ and V_{ov}. However, to derive this expression we used $g_m = \frac{2I_D}{V_{ov}}$ which is based on the square-law. For a real MOSFET, if we compute V_{ov} and $\frac{2I_D}{g_m}$ they will not be equal. Let's define a new parameter called V-star (V^*) which is calculated from actual simulation data using the formula $V^* = \frac{2I_D}{g_m} \leftrightarrow g_m = \frac{2I_D}{V^*}$

For a square-law device, $V^* = V_{ov}$, However, for a real MOSFET they are not equal. The actual gain is now given by: $|Av| \approx \frac{2V_A}{V^*}$

The lower the V^* the higher the gain, but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200\ mV$.

2) We want to design CS and cascode amplifiers with the parameters below.

Parameter	0.13um CMOS	0.18um CMOS
L [3]	0.5 μm	0.5 μm
V^*	**200 mV**	**200 mV**
Supply	**1.2 V**	**1.8 V**
Current consumption	**20 μA**	**20 μA**

$$Put\ L = 0.5\ \mu m$$

Again, I decided – in this lab – to use the **umc13mmrf** PDK (Process Design Kit), which is a 130nm process technology, instead of the **ee214b.sp** file. The reason is a matter of preference.

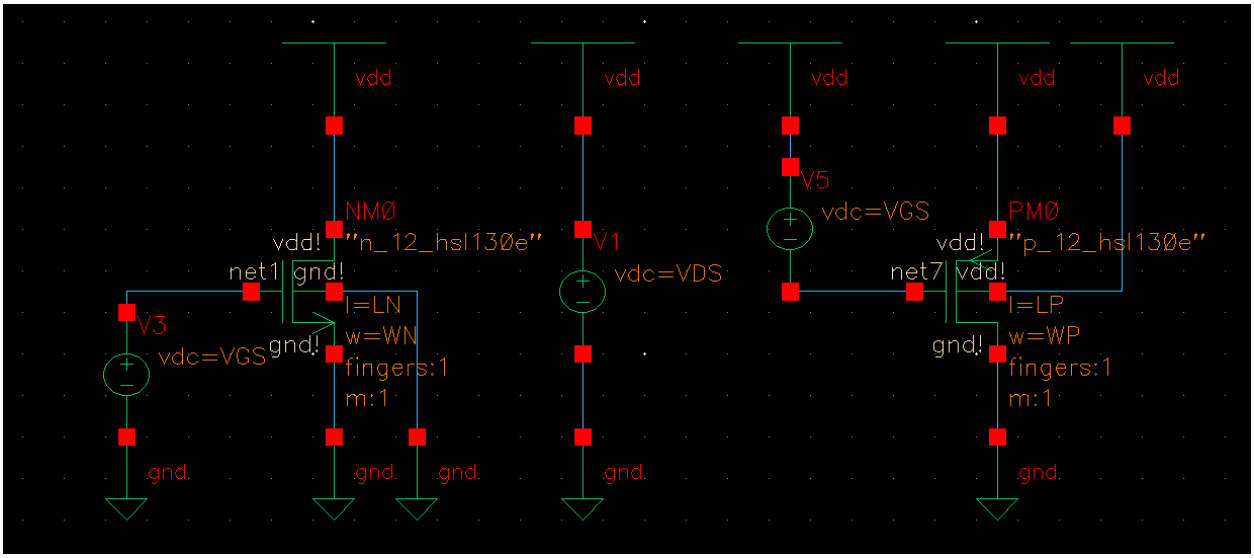
[1] *COLOR CODES:* Orange (steps) -- Blue (output) -- Green (comments / observations) – White (equations).

[2] **Cadence Virtuoso** software on a Linux® Debian 8 Virtual Machine is used.

[3] A relatively small **L** is necessary for Part 3 to make sure that **Cgd** is not negligible compared to **Cgs**.

3) The remaining variable in the design is to calculate W. Since the square-law is not accurate, we cannot use it to determine the sizing. Instead, we will use a sizing chart generated from simulation.

Create a test-bench for NMOS and PMOS characterization (we will use the PMOS later in Part 2 of this lab). Use $\boldsymbol{W}$ = 10 $\boldsymbol{\mu m}$ (we will understand why shortly) and $\boldsymbol{L}$ = 0.5 $\boldsymbol{\mu m}$ (the same L selected before).



4) Sweep VGS from $0\ to\ \approx\ V_{TH}\ + 0.4V$ with 10mV step. Set $V_{DS}\ =\ V_{DD}/2$.

$$V_{TH}\ + 0.4V\ \approx\ 0.87\ V\ [let\ it\ be\ 1V], \qquad V_{DD} = 1.2\ V, \qquad V_{DS} = 0.6\ V$$

5) We want to compare $\boldsymbol{V^*\ = 2I_D/g_m}$ and $\boldsymbol{V_{ov}\ =\ V_{GS} - V_{TH}}$ by plotting them overlaid. Use the calculator to create expressions for $\boldsymbol{V^*}$ and $\boldsymbol{V_{ov}}$. You can save the expressions to reuse them later.

The Blue Curve is $\boldsymbol{V_{ov}\ =\ V_{GS} - V_{TH}}$

```
(v("NM0:vgs" ?result "dc") - v("NM0:vth" ?result "dc"))
```

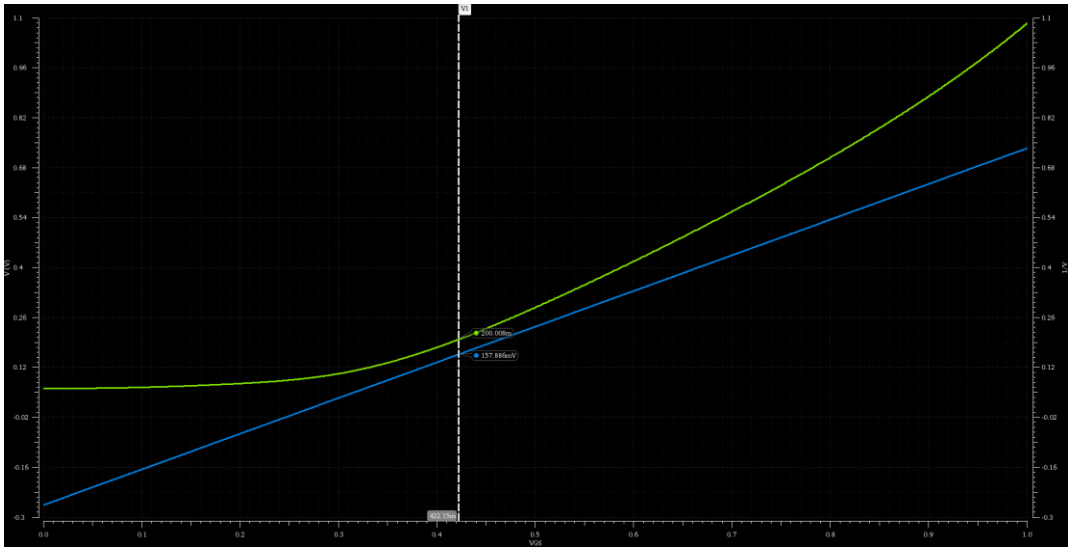
The Green Curve is $\boldsymbol{V^*\ = 2I_D/g_m}$

```
(2 / (getData("NM0:gm" ?result "dc") / getData("NM0:id" ?result "dc")))
```

The difference is that $\boldsymbol{V^*}$ takes a lot of factors into consideration, unlike the simple $\boldsymbol{V_{ov}}$ from square law.

6) Plot $\boldsymbol{V^*}$ and $\boldsymbol{V_{ov}}$ overlaid vs. $\boldsymbol{V_{GS}}$. Make sure the y-axis of both curves has the same range. You will notice that at the beginning of the strong inversion region, $\boldsymbol{V^*}$ and $\boldsymbol{V_{ov}}$ are relatively close to each other (i.e., square-law is relatively valid). For deep strong inversion (large $\boldsymbol{V_{ov}}$: velocity saturation and mobility degradation) or weak inversion (near-threshold and sub-threshold operation) the behavior is quite far from the square-law.

After adjusting the y-axes ranges to be the same (-300mV to 1.1V) and x-axis is not changed (0 to 1V).



7) An often used sweet-spot that provides good compromise between different trade-offs is $\boldsymbol{V^*} = 200\ mV$. On the $\boldsymbol{V^*}$ and $\boldsymbol{V_{ov}}$ chart locate the point at which $\boldsymbol{V^*} = 200\ mV$. Find the corresponding $\boldsymbol{V_{ovQ}}$ and $\boldsymbol{V_{GSQ}}$.

$$\boldsymbol{V^* = V_Q^* = 200\ mV}$$

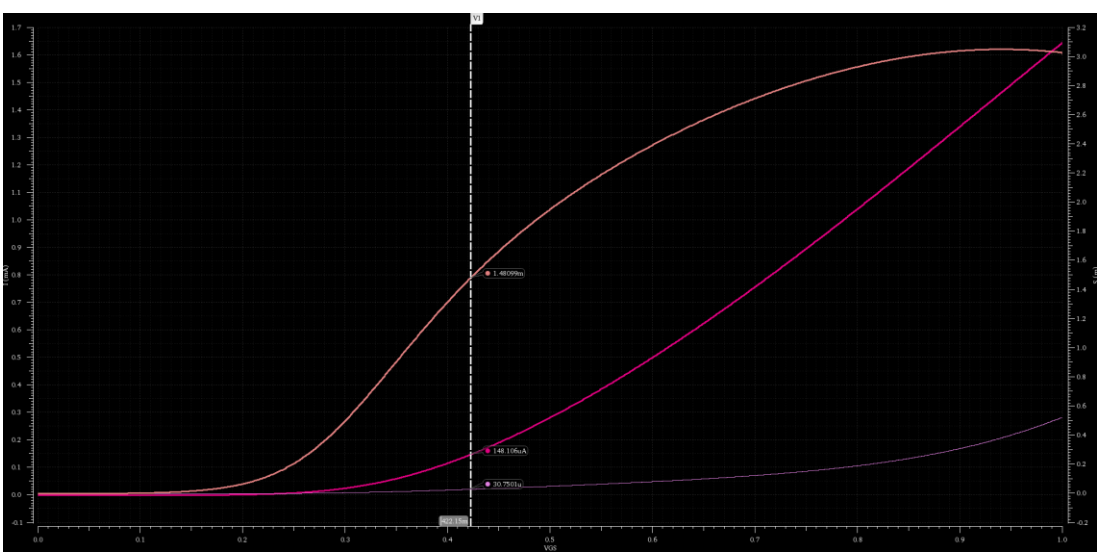
$$\boldsymbol{V_{ov} = V_{ov_Q} = 157.886\ mV}$$

It is noted that $\boldsymbol{V_{GS} = V_{GS_Q} = 422.15\ mV}$

8) Plot $I_D,\ g_m, and\ g_{ds}$ vs. V_{GS}. Find their values at V_{GS_Q}. Let's name these values $I_{DX}, g_{mX}, and\ g_{dsX}$.

$\boldsymbol{I_{DX} = 148.1\ \mu A}$
$\boldsymbol{g_{mX} = 1.481\ mS}$
$\boldsymbol{g_{dsX} = 30.75\ \mu S}$
$\boldsymbol{r_o = 32.52\ k\Omega}$

ID_n	
Vstar_n	
Vov_n	
id_n	148.1u
gm_n	1.481m
gds_n	30.75u
ro_n	32.52k



9) Now back to the assumption that we made that $\boldsymbol{W}$ = 10$\boldsymbol{\mu m}$. This is not the actual value that we will use for our design. But the good news is that $\boldsymbol{I_D}$ is always proportional to $\boldsymbol{W}$ irrespective of the operating region and the model of the MOSFET (regardless square-law is valid or no). Thus, we can use ratio and proportion (cross-multiplication) to determine the correct width at which the current will be $\boldsymbol{I_{DQ} = 20\mu A}$ as given in the specs. Calculate $\boldsymbol{W}$ as shown below.

W	$\boldsymbol{I_D}$
10 µm	I_{DX} @ V_Q^* (from the chart)
?	$I_{DQ} = 20\ \mu A$ (from the specs)

$$W = \frac{10\ \mu m \times 20\ \mu A}{I_{DX}\ @\ V_Q^*\ \text{(from the chart)}}$$

$$W = \frac{10\ \mu m \times 20\ \mu A}{\mathbf{148.1\ \mu A}} = 1.3505\ \mu m \approx \mathbf{1.4\ \mu m}$$

10) Now we are almost done with the design of the amplifier. Note that $\boldsymbol{g_m}$ is also proportional to $\boldsymbol{W}$ as long as $\boldsymbol{V_{ov}}$ is constant. On the other hand, $\boldsymbol{r_o = 1/g_{ds}}$ is inversely proportional to $\boldsymbol{W}$ $(\boldsymbol{I_D})$ as long as L is constant. Before leaving this part, calculate $\boldsymbol{g_{mQ}}$ and $\boldsymbol{g_{dsQ}}$ using ratio and proportion (cross multiplication).

$\boldsymbol{g_{mQ}}$	W @ constant $\boldsymbol{V_{ov}}$
$\mathbf{1.481\ mS}$	$10\ \mu m$
?	$1.4\ \mu m$

$\boldsymbol{g_{dsQ}}$	W $(\boldsymbol{I_D})$
$\mathbf{30.75\ \mu S}$	$10\ \mu m$
?	$1.4\ \mu m$

$$\boldsymbol{g_{mQ}} = \frac{\mathbf{1.481\ mS} \times 1.4\ \mu m}{10\ \mu m} = 0.20734\ mS$$

$$\boldsymbol{g_{dsQ}} = \frac{\mathbf{4.5\ \mu S} \times 1.4\ \mu m}{10\ \mu m} = 4.305\ \mu S$$

$$r_o = \frac{1}{4.305\ \mu S} \approx 232.288\ k\Omega$$

<u>Note:</u> $g_m = 1.481\ mS \rightarrow\ g_m = 0.20734\ mS$
Directly proportional to W (decreased)

$r_o = 32.52\ k\Omega \rightarrow r_o \approx 232.288\ k\Omega$
Inversely proportional to W (increased)

REVIEW ON OUR FINAL PARAMETERS:

$$\therefore L = 0.5\ \mu m, W = 1.4\ \mu m, V_{DS} = 0.6\ V,$$

$$\boldsymbol{V^* = V_Q^* = 200\ mV}$$
$$\boldsymbol{V_{ov} = V_{ov_Q} = 157.886\ mV}$$
$$\boldsymbol{V_{GS} = V_{GS_Q} = 422.15\ mV}$$

$$\therefore g_m = 207.34\ \mu S, \qquad g_{ds} = 4.305\ \mu S, \qquad r_o = 232.288\ k\Omega$$

Simulator Results:

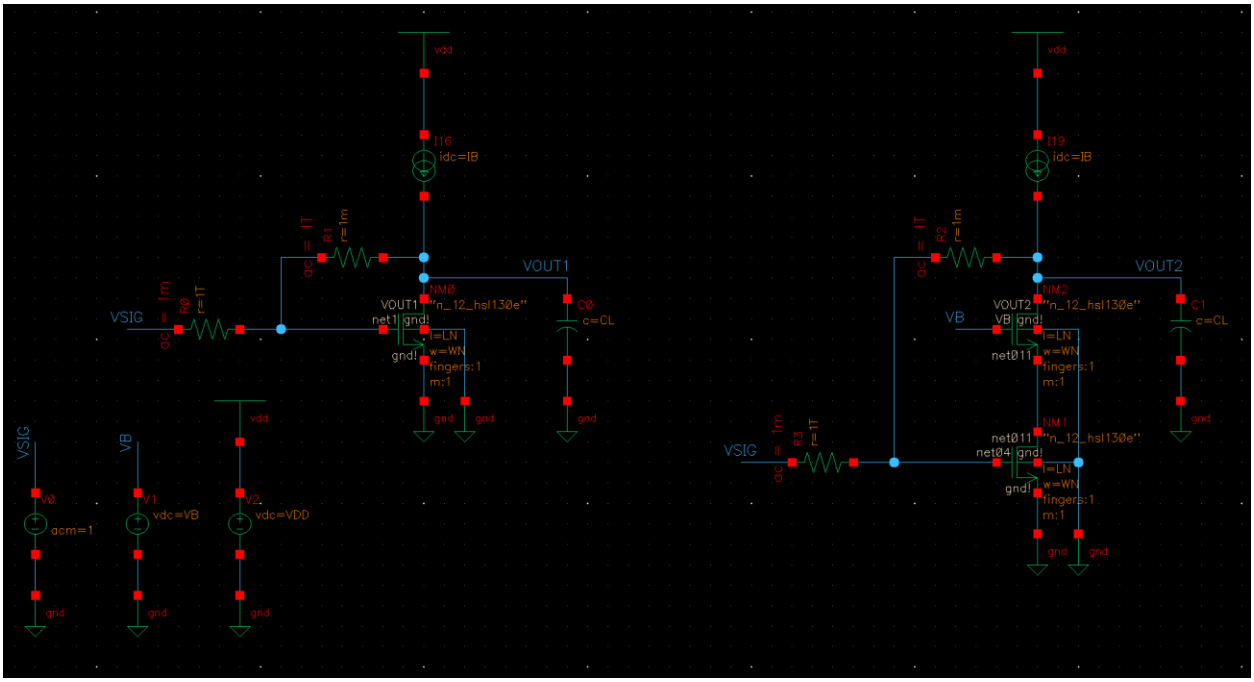
ID_n	
Vstar_n	
Vov_n	
id_n	19u
gm_n	195.4u
gds_n	3.928u
ro_n	254.6k

END OF SIZING AND DESIGN

Part 2: Cascode for Gain

1. OP Analysis

1) Create a new cell and schematic. Construct the circuit shown below. Use $I_B = 20\mu A$, $L = 0.5\mu m$, W as selected in Part 1 ($W = 1.4\mu m$), and $C_L = 1pF$.



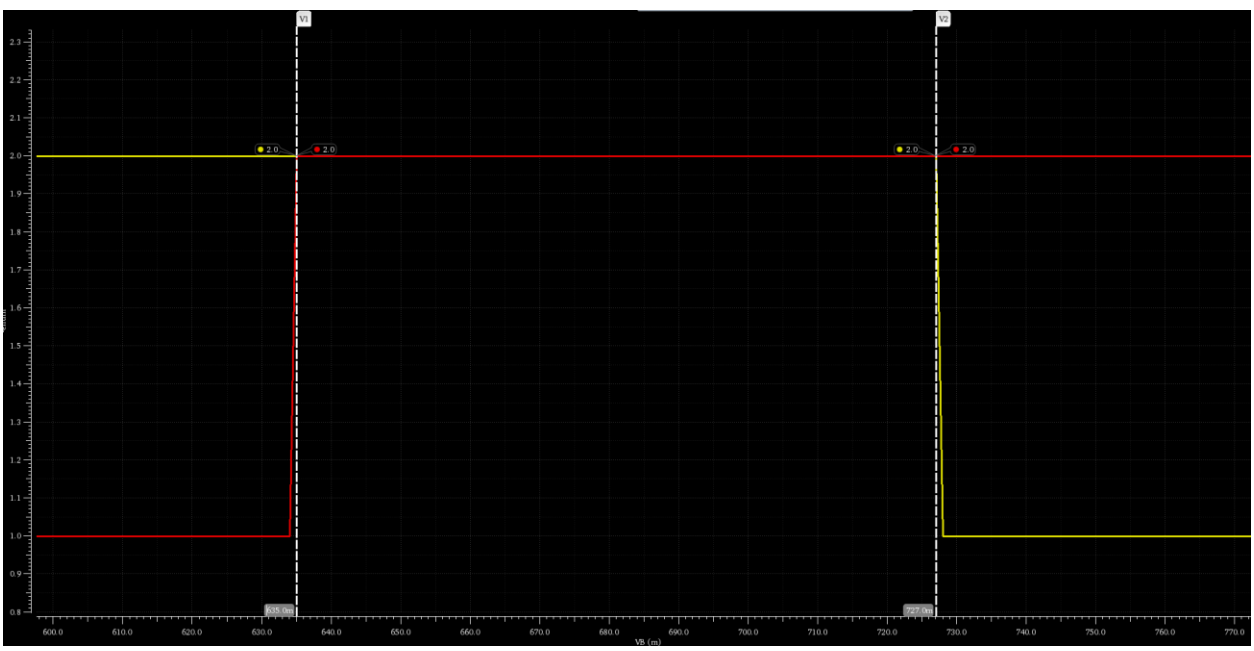
2) Choose V_B (the cascode device bias voltage) such that M2 (NM1 in schematic) has $V_{DS} \approx V^* + 100\ mV$ (you may sweep V_B and plot V_{DS} vs. V_B to help you choose a good value for V_B).

$$V_{DS,NM1} \approx V^* + 100\ mV \approx 200\ mV + 100\ mV = 300\ mV$$

$$V_B = V_{GS,NM2} + V_{DS,NM1} = \sqrt{\frac{2(20\mu A)}{\beta}} + V_{th} + 300\ mV \approx 600\ mv$$

By trying multiple sweep values for VB, we find that there are some value that result in a breakdown region, so we have to be careful in our sweep start voltage (just 0:1m:1V will result in ERROR).

By trial and error, we sweep VB from 0.5:1m:1V then $sweep\ \boldsymbol{V_B}\ and\ plot\ \boldsymbol{V_{DS,M2}}\ \boldsymbol{vs.V_B}$



Both NM1, NM2 are in saturation when VB = 635mV to 727mV

These ranges can be deduced analytically or by intuition:

$$\boldsymbol{V_B = V_{GS,NM2} + V_{DS,NM1}}$$

$$\begin{aligned}
&for\ NM1\ (CS)\ to\ be\ in\ saturation: \boldsymbol{V_{DS,NM1} \geq V^*}\\
&\boldsymbol{V_B - V_{GS,NM2} \geq V^*}\\
&\boldsymbol{V_B \geq V^* + (V^* + V_{th})}\\
&\boldsymbol{V_B \geq 2V^* + V_{th}}\\
&V_{B,min} = 2V^* + V_{th} = 2(200m) + 270m = 670mV
\end{aligned}$$

A more accurate result will be by using $\boldsymbol{V_{D,sat}}$

$$V_{B,min} = 2\boldsymbol{V_{D,sat}} + V_{th} = 2(175m) + 270m = 620mV$$

$$\begin{aligned}
&for\ NM2\ (cascode) to\ be\ in\ saturation: \boldsymbol{V_{DS,NM2} \geq V^*}\\
&\boldsymbol{V_{out_{dc}} - V_{DS,NM1} \geq V^*}\\
&\boldsymbol{V_{out_{dc}} - (V_B - V_{GS}) \geq V^*}\\
&\boldsymbol{V_{out_{dc}} - V_B + V_{GS} \geq V^*}\\
&\boldsymbol{V_B \leq V_{out_{dc}} + V_{GS} - V^*}\\
&V_{B,MAX} = V_{out_{dc}} + \boldsymbol{V^* + V_{th}} - V^* = \boldsymbol{V_{out_{dc}} + V_{th} = V_{GS} + V_{th}}\\
&= \boldsymbol{V^* + 2V_{th}} = 200m + 2(270m) = 740mV
\end{aligned}$$

A more accurate result will be by using $\boldsymbol{V_{D,sat}}$

$$V_{B,min} = \boldsymbol{V_{D,sat} + 2V_{th}} = 175m + 2(270m) = 715mV$$

3) We need to bias transistors in saturation; however, the output node is a high impedance node; thus, it is difficult to control its DC voltage. As a workaround in simulation, we use a feedback loop and resistors with different resistances in DC/AC to change the circuit connections in DC/AC simulations (use the AC property in ideal_resistor). The input transistor is diode connected for DC simulation (always in saturation), while in AC simulation the feedback is disconnected, and the AC input source is connected. Set the feedback resistance $\boldsymbol{1\ m\Omega\ DC}$ and $\boldsymbol{1\ T\Omega\ AC}$ and set the source resistance oppositely.

We will study how to do biasing practically later in this course inshaAllah.

$$\boldsymbol{Choose\ V_B = 700mV}$$

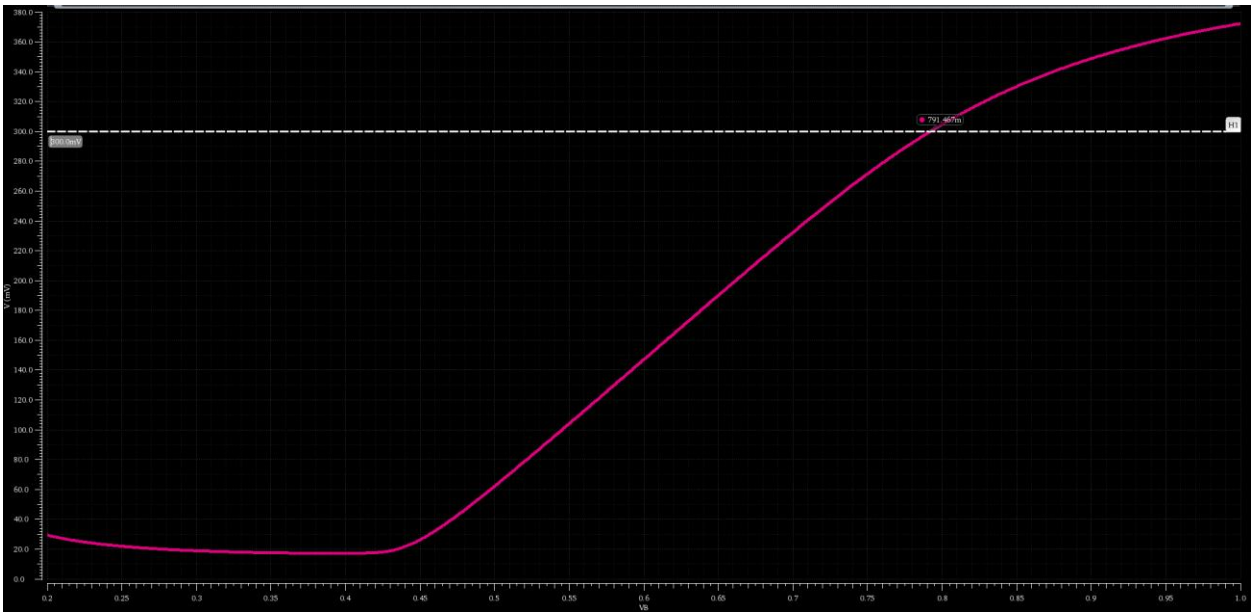
4) Simulate the DC OP point of the above CS and cascode amplifiers. Report a snapshot showing the following parameters for M1, M2 and M3 in addition to DC node voltages clearly annotated.

Parameter \ Transistor	M1 (NM0) (CS only)	M2 (NM1) (CS in cascode)	M3 (NM2) (CG in cascode)
ID	20.03u	20u	20u
VGS	431m	437m	467.2m
VDS	431m	232.8m	204.3m
VTH	269.6m	269.6m	298.4m (body effect)
VDSAT	168.9m	173.4m	178.9m
GM	197.8u	188.8u	182.5u
GDS	4.561u	10.81u	17.54u
GMB	25.67u	24.65u	20.38u
CDB	0.7137f	0.8002f	0.7325f
CGD	0.4128f	0.4663f	0.5089f
CGS	5.344f	5.368f	5.342f
CSB	1.5f	1.497f	1.271f
Region	2 = saturation	2 = saturation	2 = saturation

Output	Nominal
getData("NM0:id" ?result "dcOp")	20.03u
getData("NM1:id" ?result "dcOp")	20u
getData("NM2:id" ?result "dcOp")	20u
v("NM0:vgs" ?result "dcOp")	431m
v("NM1:vgs" ?result "dcOp")	437m
v("NM2:vgs" ?result "dcOp")	467.2m
v("NM0:vds" ?result "dcOp")	431m
v("NM1:vds" ?result "dcOp")	232.8m
v("NM2:vds" ?result "dcOp")	204.3m
v("NM0:vth" ?result "dcOp")	269.6m
v("NM1:vth" ?result "dcOp")	269.6m
v("NM2:vth" ?result "dcOp")	298.4m
v("NM0:vdsat" ?result "dcOp")	168.9m
v("NM1:vdsat" ?result "dcOp")	173.4m
v("NM2:vdsat" ?result "dcOp")	178.9m
getData("NM0:gm" ?result "dcOp")	197.8u
getData("NM1:gm" ?result "dcOp")	188.8u
getData("NM2:gm" ?result "dcOp")	182.5u
getData("NM0:gds" ?result "dcOp")	4.561u
getData("NM1:gds" ?result "dcOp")	10.81u
getData("NM2:gds" ?result "dcOp")	17.54u
getData("NM0:gmb" ?result "dcOp")	25.67u
getData("NM1:gmb" ?result "dcOp")	24.65u
getData("NM2:gmb" ?result "dcOp")	20.38u

getData("NM0:cdb" ?result "dcOp")	-713.7a
getData("NM1:cdb" ?result "dcOp")	-800.2a
getData("NM2:cdb" ?result "dcOp")	-732.5a
getData("NM0:cgd" ?result "dcOp")	-412.8a
getData("NM1:cgd" ?result "dcOp")	-466.3a
getData("NM2:cgd" ?result "dcOp")	-508.9a
getData("NM0:cgs" ?result "dcOp")	-5.344f
getData("NM1:cgs" ?result "dcOp")	-5.368f
getData("NM2:cgs" ?result "dcOp")	-5.342f
getData("NM0:csb" ?result "dcOp")	-1.5f
getData("NM1:csb" ?result "dcOp")	-1.497f
getData("NM2:csb" ?result "dcOp")	-1.271f
getData("NM0:region" ?result "dcOp")	2
getData("NM1:region" ?result "dcOp")	2
getData("NM2:region" ?result "dcOp")	2

"vdsat" is the minimum drain-source voltage required to bias the transistor in saturation. It is equal to $\boldsymbol{V_{ov}}$ for a square-law device. It is also referred to as "vdss" (drain-source saturation voltage).



$$\boldsymbol{V_B = 791.467m} \xrightarrow{for} \boldsymbol{V_{DS} = V^* + 100m}$$

5) Check that all transistors operate in saturation.

To make sure all transistors operate in saturation we found that VB has to be within a certain range:

By simulation VB = **635mV** to **727mV**
By Hand Analysis VB = **620mV** to **715mV**

As required in lab instruction ((Choose $\boldsymbol{V_B}$ such that M2 (NM1) has $\boldsymbol{V_{DS} \approx V^* + 100\ mV}$)). To achieve this constraint **VB will have to be about 790mV to 800mV.** This will certainly knock M3 (NM2 – Cascode Transistor) out of saturation into triode.

So it was decided to use **VB=700mV** and make sure all transistors operate in saturation, but **VDS = 232.8m**, which means that **VDS is not V*+100m**=211.8+100=311mV instead **it is V*+20mV only**.

6) Do all transistors have the same V_{th}? Why?

NM0: 269.6m (CS only)

NM1: 269.6m (CS in cascode)

NM2: 298.4m (CG in cascode transistor) → higher $\boldsymbol{v_{th}}$ due to <u>**body effect**</u>.

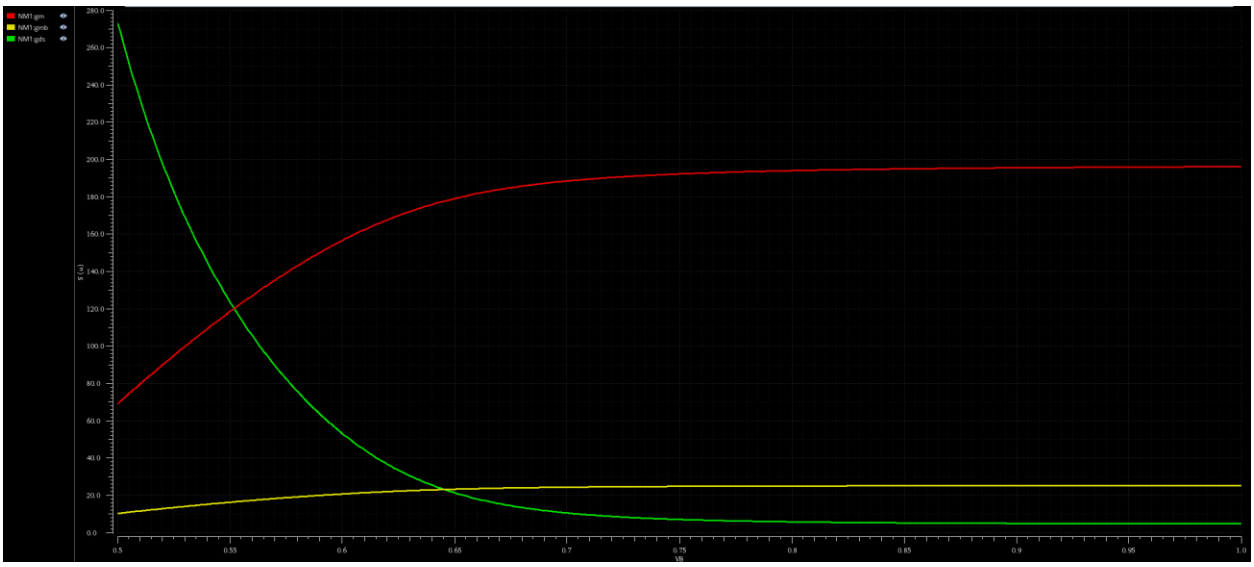
$$V_{th} = V_{th0} + \gamma \left(\sqrt{2\phi_f + 2V_{SB}} - \sqrt{|2\phi_f|} \right)$$

7) What is the relation (≪, ≫) between g_m and g_{ds}?

$$g_m >> g_{ds}$$

8) What is the relation (≪, ≫) between g_m and g_{mb}?

$$g_m >> g_{mb}$$

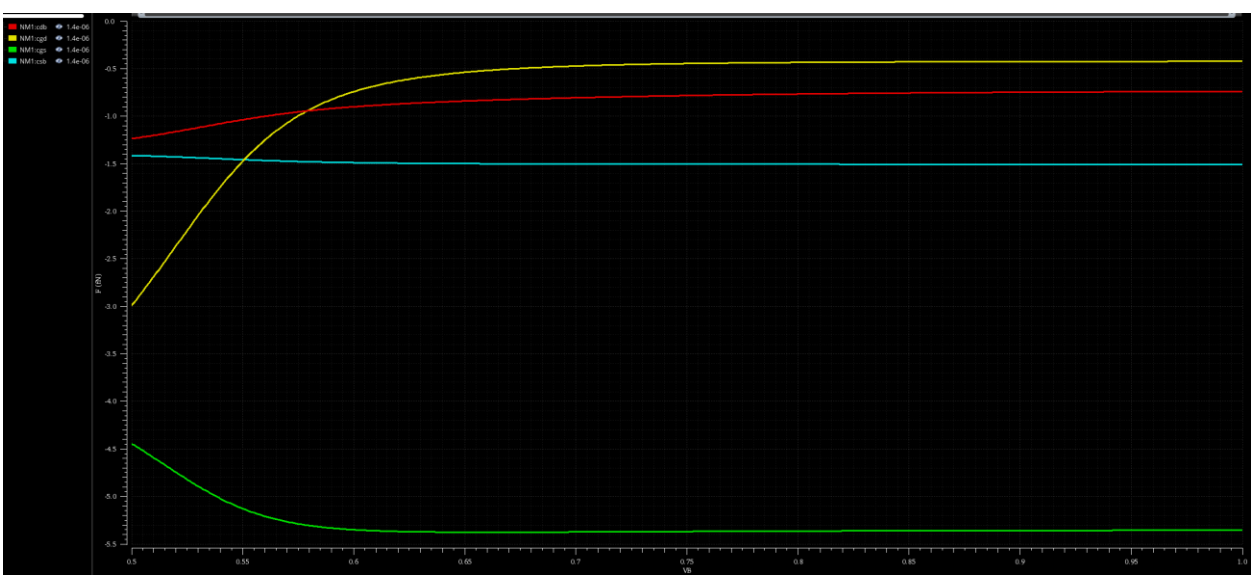


9) What is the relation (≪, ≫) between c_{gs} and c_{gd}?

$$c_{gs} \gg c_{gd}$$

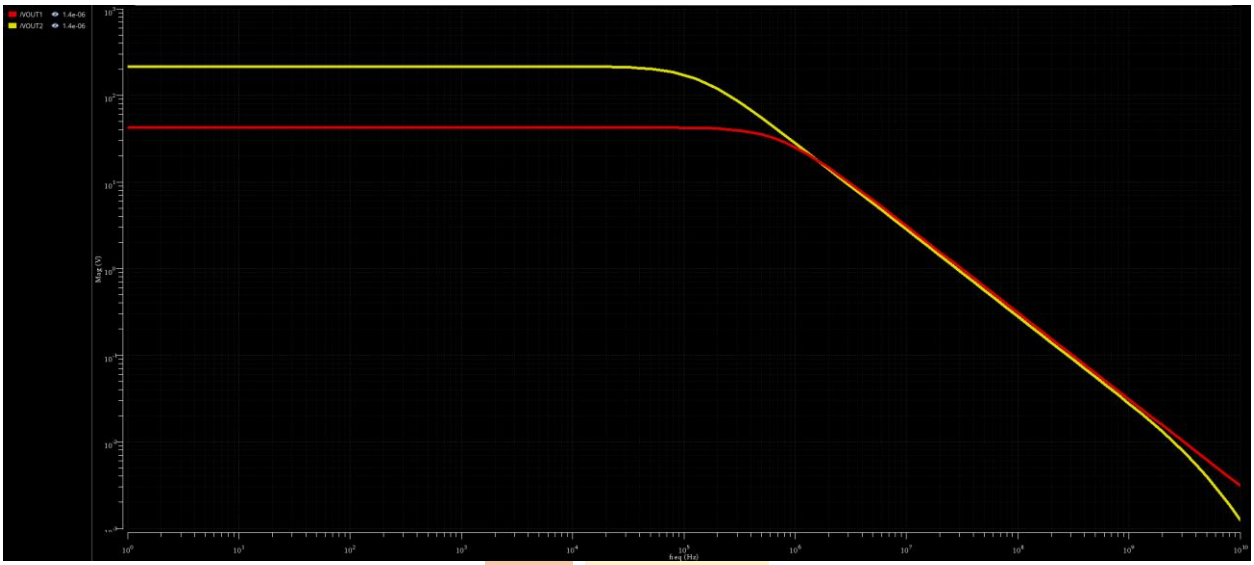
10) What is the relation (≪, ≫) between csb and cdb?

$$c_{sb} > c_{db}$$



2. AC Analysis

1) Create a new simulation configuration. Perform AC analysis (**1Hz: 10GHz, logarithmic, 10points/decade**) to simulate gain and bandwidth.



Red: CS, Yellow: cascode

2) Use calculator to create expressions for circuit parameters (DC gain, BW, GBW, and UGF) and export them to adexl.

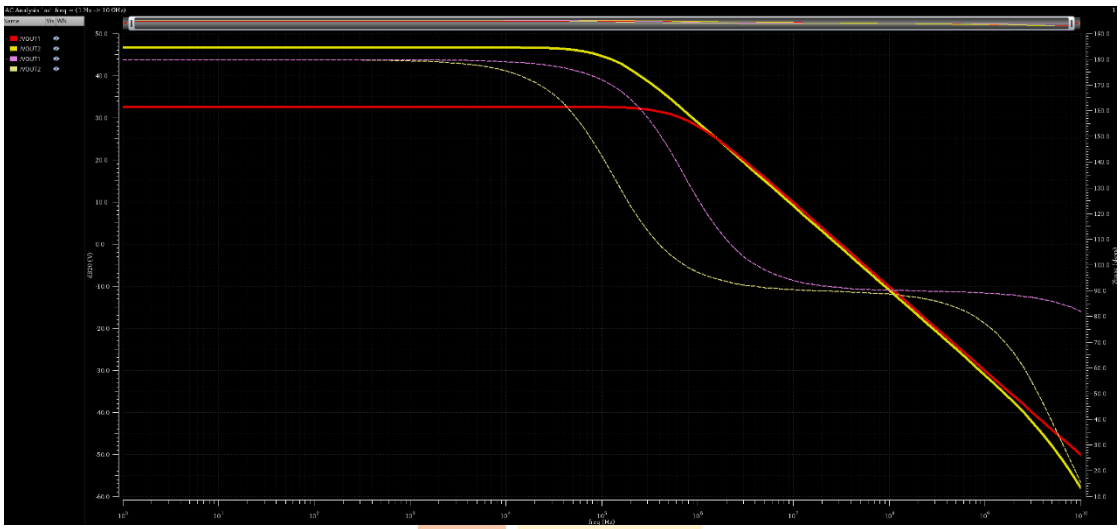
➔ **Cadence Hint:** Use the following expressions in the calculator, and send them to adexl to quickly calculate circuit parameters.

<u>Commands:</u>

```
mag(v("/VOUT1" ?result "ac"))
db20(v("/VOUT1" ?result "ac"))
ymax(db20(v("/VOUT1" ?result "ac")))
bandwidth(mag(v("/VOUT1" ?result "ac")) 3 "low")
gainBwProd(mag(v("/VOUT1" ?result "ac")))
unityGainFreq(mag(v("/VOUT1" ?result "ac")))
```

mag(v("/VOUT1" ?result "ac"))	
db20(v("/VOUT1" ?result "ac"))	
DCgain_CS	32.74
BW_CS	725.5k
GBW_CS	31.54M
UGF_CS	31.48M
mag(v("/VOUT2" ?result "ac"))	
db20(v("/VOUT2" ?result "ac"))	
DCgain_cascode	46.83
BW_cascode	130.1k
GBW_cascode	28.65M
UGF_cascode	28.96M

3) Report the Bode plot (magnitude) of CS and cascode <u>appended on the same plot.</u>



Red: CS, Yellow: cascode

4) Using small signal parameters from OP simulation, perform <u>hand analysis</u> to calculate DC gain, BW, and GBW of both circuits.

DC GAIN	
CS	Cascode (with body effect on NM2)
$\boldsymbol{DC\ Gain_{CS} = A_{v_0} = -g_m r_o}$	$\boldsymbol{DC\ Gain_{Cascode} = A_{v_0} = -g_{m1}\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})]}$
$A_{v_0} = -g_m r_o = -\frac{g_m}{g_{ds}}$ $= \frac{-197.7\mu}{4.556\mu} = -43.3933$ $20\log(43.3933) = 32.74dB$	$A_{v_0} = -g_m R_{LFD,NM2}$ $= -g_{m,NM1}\left(r_{o,NM2}([g_{m,NM2} + g_{mb,NM2}]\, r_{o,NM1})\right)$ $= -\frac{g_{m,NM1}[g_{m,NM2} + g_{mb,NM2}]}{g_{ds,NM1} g_{ds,NM2}}$ $= \frac{-188.8\mu \times [182.5\mu + 20.38\mu]}{10.81\mu \times 17.54\mu} = -202.016$ $20\log(202.016) = 46.1077dB$

BW	
CS	Cascode
$\boldsymbol{A_v(s) = -\frac{g_m r_o}{1 + s\, r_o C_L}}$ $\omega_p = \frac{1}{RC} = \frac{1}{r_o C_L}$ $\rightarrow\ \boldsymbol{f_p = \frac{1}{2\pi\, r_o C_L} = \frac{g_{ds}}{2\pi C_L}}$	$\boldsymbol{A_v(s) = \frac{-g_{m1}\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})]}{1 + s\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})]C_L}}$ $\omega_p = \frac{1}{RC} = \frac{1}{[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})]C_L}$ $= \frac{g_{ds2}}{(1 + [g_{m2} + g_{mb2}]r_{o1})C_L} \rightarrow \boldsymbol{f_p = \frac{g_{ds2}}{2\pi\, C_L(1 + \frac{g_{m2} + g_{mb2}}{g_{ds1}})}}$
$BW = \frac{g_{ds}}{2\pi C_L} = \frac{4.556\mu}{2\pi(1p)}$ $= 725.11kHz$	$BW = \frac{17.54\mu}{2\pi(1p)(1 + \frac{182.5\mu + 20.38\mu}{10.81\mu})} = 141.218kHz$

GBW	
CS	Cascode
$\boldsymbol{GBW} = \lvert A_{v_0} \rvert \times f_p$ $= -g_m r_o \times \frac{1}{2\pi\, r_o C_L} = \boldsymbol{\frac{g_m}{2\pi C_L}}$	$\boldsymbol{GBW} = \lvert A_{v_0} \rvert \times f_p =$ $\frac{g_{m1}[g_{m2}+g_{mb2}]}{g_{ds1} g_{ds2}} \times \frac{g_{ds2}}{2\pi\, C_L \left(1+\frac{g_{m2}+g_{mb2}}{g_{ds1}}\right)} =$ $\boldsymbol{\frac{g_{m1}[g_{m2}+g_{mb2}]}{2\pi g_{ds1}\, C_L (1+\frac{g_{m2}+g_{mb2}}{g_{ds1}})}}$
$GBW = \frac{g_m}{2\pi C_L} = \frac{197.7\mu}{2\pi(1p)}$ $= 31.465MHz$	$GBW = \frac{g_{m1}[g_{m2}+g_{mb2}]}{2\pi g_{ds1}\, C_L (1+\frac{g_{m2}+g_{mb2}}{g_{ds1}})} = 28.528MHz$

UGF	
CS	Cascode
$let:\ \lvert A_v(s) \rvert = -\frac{g_m r_o}{1+\boldsymbol{s}\, r_o C_L} = 1$ $1+\boldsymbol{s}\, r_o C_L = -g_m r_o$ $s\, r_o C_L = -g_m r_o - 1$ $\boldsymbol{s} = -\omega_u = -2\pi f_u = \frac{-g_m r_o - 1}{r_o C_L}$ $\boldsymbol{UGF = f_u = \frac{g_m r_o + 1}{2\pi\, r_o C_L} \approx \frac{g_m r_o}{2\pi\, r_o C_L} \approx \frac{g_m}{2\pi C_L}}$ $\boldsymbol{\approx GBW}$	$1 = \frac{-g_{m1}\,[r_{o2}(1+[g_{m2}+g_{mb2}]r_{o1})]}{1+\boldsymbol{s}\,[r_{o2}(1+[g_{m2}+g_{mb2}]r_{o1})]C_L}$ $1+\boldsymbol{s}\, u C_L = -g_{m1} u$ $\boldsymbol{s} = -\omega_u = -2\pi f_u = \frac{-g_{m1} u - 1}{u C_L}$ $\boldsymbol{UGF = f_u = \frac{g_{m1} u + 1}{2\pi u C_L} \approx \frac{g_{m1} u}{2\pi u C_L} \approx \frac{g_{m1}}{2\pi C_L}}$
$GBW = \frac{g_m}{2\pi C_L} = \frac{197.7\mu}{2\pi(1p)} = 31.465MHz$	$UGF = \frac{g_{m1}}{2\pi C_L} = \frac{197.7\mu}{2\pi(1p)} = 30.05MHz$

5) Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis.

	Simulation		Hand Analysis	
	CS	Cascode	CS	Cascode
DC gain	32.74dB	46.83dB	32.74dB	46.1077dB
BW	725.5k	130.1k	725.11k	141.218k
GBW	31.54M	28.65M	31.465M	28.528M
UGF	31.48M	28.96M	31.465M	30.05M

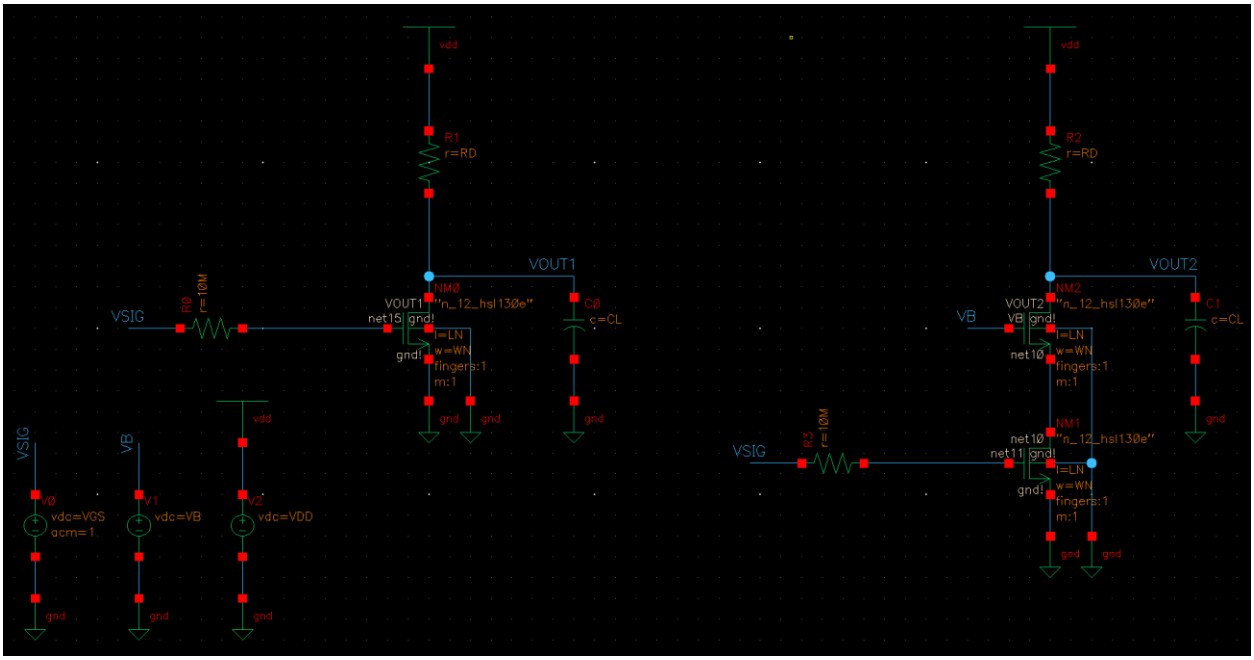
6) Comment on the results.

Simulation and hand-analysis are close. The **cascode amplifier** has much higher gain than the **common source amplifier** but much less bandwidth, so the gain-bandwidth product remains constant.

Part 3 [Optional]: Cascode for Bandwidth

1. OP Analysis

1) Create a new cell and schematic. Copy the old schematic instances to the new one. Make the following modifications: remove the feedback resistance, set $\boldsymbol{C_L = 1fF}$, replace the current source with a resistor $\boldsymbol{R_D}$, and make the signal source resistance = $\boldsymbol{10M\Omega}$ (remove the AC value). Set VGS of the input device as calculated in Part 1.



2) Calculate $\boldsymbol{R_D}$ analytically such that the voltage drop on it is ≈ $\boldsymbol{V_{DD}/2}$ (the current remains roughly the same as in Part 2 because we are using the same VGS).

$$\boldsymbol{V_{GS} = V_{GS_Q} = 422.15\ mV}$$

$$\boldsymbol{V_{R_D} \approx \frac{V_{DD}}{2} \approx 0.6V}$$

$$I_D R_D \approx \boldsymbol{0.6V}$$

$$R_D = \frac{0.6V}{20\mu A} = 30k\Omega$$

3) Note that the DC voltage of the output node is set by the resistance ($\boldsymbol{R_D}$); thus, we don't need a feedback loop as in the previous case.

4) Simulate the DC OP point of the new CS and cascode amplifiers. Add monitors and report a snapshot showing the following parameters for M1, M2 and M3 in addition to DC node voltages clearly annotated.

We will use the same $V_B = 700mV$ as part 2 for the Cascode Amplifier.

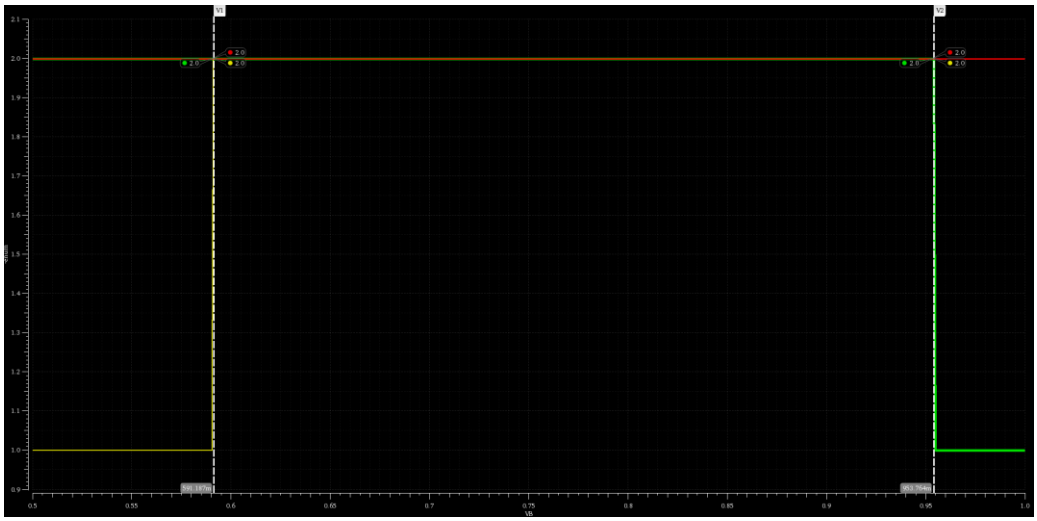
Parameter \ Transistor	M1 (NM0) (CS only)	M2 (NM1) (CS in cascode)	M3 (NM2) (CG in cascode)
ID	19.11u	17.43u	17.43u
VGS	422.2m	422.2m	447.5m
VDS	626.7m	252.5m	424.7m
VTH	269.6m	269.6m	300.6m (body effect)
VDSAT	162.4m	162.4m	162.6m
GM	196.3u	179.6u	184.7u
GDS	3.908u	7.053u	4.164u
GMB	25.37u	23.38u	20.16u
CDB	0.6613f	0.7865f	0.6521f
CGD	0.4078f	0.4439f	0.4124f
CGS	5.307f	5.314f	5.262f
CSB	1.498f	1.494f	1.257f
Region	2 = saturation	2 = saturation	2 = saturation

Output	Nominal
getData("NM0:id" ?result "dcOp")	19.11u
getData("NM1:id" ?result "dcOp")	17.43u
getData("NM2:id" ?result "dcOp")	17.43u
v("NM0:vgs" ?result "dcOp")	422.2m
v("NM1:vgs" ?result "dcOp")	422.2m
v("NM2:vgs" ?result "dcOp")	447.5m
v("NM0:vds" ?result "dcOp")	626.7m
v("NM1:vds" ?result "dcOp")	252.5m
v("NM2:vds" ?result "dcOp")	424.7m
v("NM0:vth" ?result "dcOp")	269.6m
v("NM1:vth" ?result "dcOp")	269.6m
v("NM2:vth" ?result "dcOp")	300.6m
v("NM0:vdsat" ?result "dcOp")	162.4m
v("NM1:vdsat" ?result "dcOp")	162.4m
v("NM2:vdsat" ?result "dcOp")	162.6m
getData("NM0:gm" ?result "dcOp")	196.3u
getData("NM1:gm" ?result "dcOp")	179.6u
getData("NM2:gm" ?result "dcOp")	184.7u
getData("NM0:gds" ?result "dcOp")	3.908u
getData("NM1:gds" ?result "dcOp")	7.053u
getData("NM2:gds" ?result "dcOp")	4.164u
getData("NM0:gmb" ?result "dcOp")	25.37u
getData("NM1:gmb" ?result "dcOp")	23.38u
getData("NM2:gmb" ?result "dcOp")	20.16u

getData("NM0:cdb" ?result "dcOp")	-661.3a
getData("NM1:cdb" ?result "dcOp")	-786.5a
getData("NM2:cdb" ?result "dcOp")	-652.1a
getData("NM0:cgd" ?result "dcOp")	-407.8a
getData("NM1:cgd" ?result "dcOp")	-443.9a
getData("NM2:cgd" ?result "dcOp")	-412.4a
getData("NM0:cgs" ?result "dcOp")	-5.307f
getData("NM1:cgs" ?result "dcOp")	-5.314f
getData("NM2:cgs" ?result "dcOp")	-5.262f
getData("NM0:csb" ?result "dcOp")	-1.498f
getData("NM1:csb" ?result "dcOp")	-1.494f
getData("NM2:csb" ?result "dcOp")	-1.257f
getData("NM0:region" ?result "dcOp")	2
getData("NM1:region" ?result "dcOp")	2
getData("NM2:region" ?result "dcOp")	2

5) Check that all transistors operate in saturation.

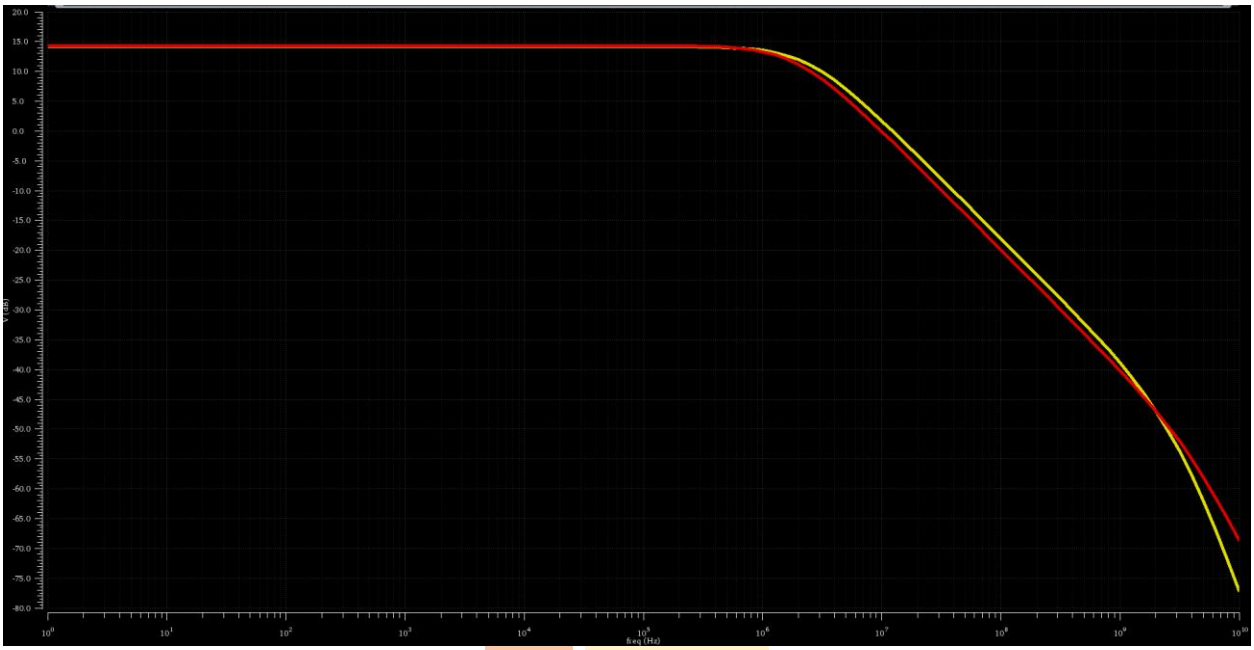
All transistors operate in saturation



All Cascode Amplifier Transistors are in saturation if V_B is in range from 591.2mV to 953mV

2. AC Analysis

1) Perform AC analysis (1Hz: 10GHz, logarithmic, 10points/decade) to simulate gain and bandwidth.



Red: CS, Yellow: cascode

2) Use calculator to create expressions for circuit parameters (DC gain, BW, GBW, and UGF) and export them to adexl.

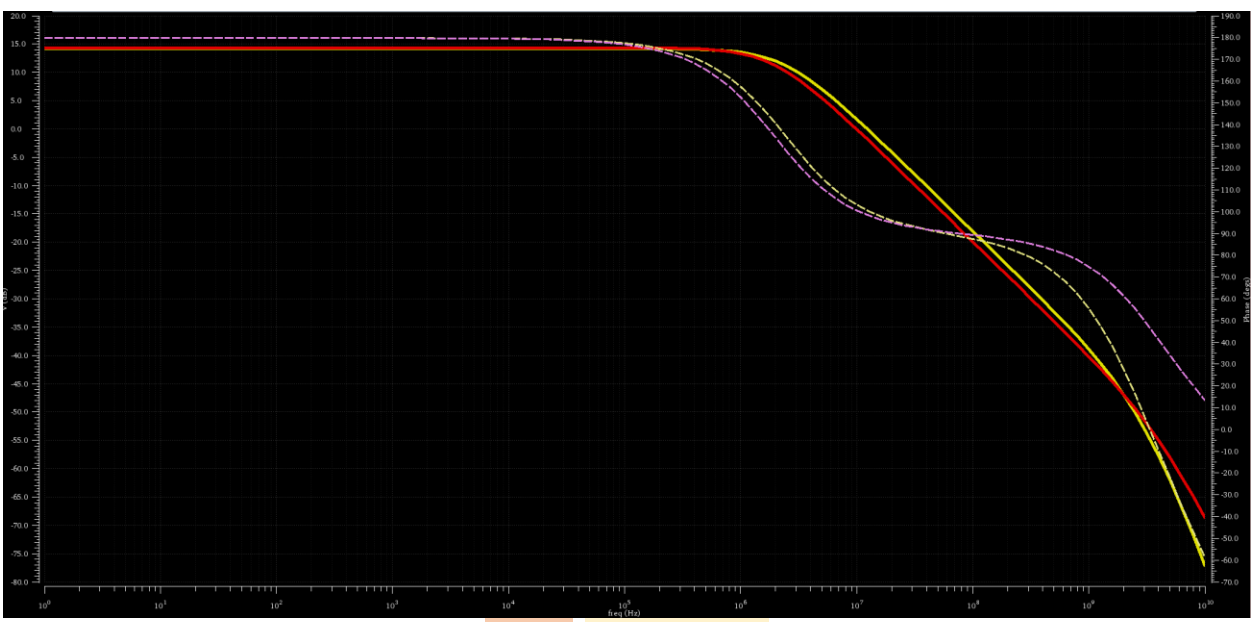
<u>Commands:</u>

```
mag(v("/VOUT1" ?result "ac"))
db20(v("/VOUT1" ?result "ac"))
ymax(db20(v("/VOUT1" ?result "ac")))
bandwidth(mag(v("/VOUT1" ?result "ac")) 3 "low")
gainBwProd(mag(v("/VOUT1" ?result "ac")))
unityGainFreq(mag(v("/VOUT1" ?result "ac")))
```

mag(v("/VOUT1" ?result "ac"))	
db20(v("/VOUT1" ?result "ac"))	
DCgain_CS	14.44
BW_CS	1.933M
GBW_CS	10.21M
UGF_CS	10.01M
mag(v("/VOUT2" ?result "ac"))	
db20(v("/VOUT2" ?result "ac"))	
DCgain_cascode	14.3
BW_cascode	2.435M
GBW_cascode	12.67M
UGF_cascode	12.45M

Note that the gain of CS and cascode for BW are the same but cascode for BW has bigger BW.

3) Report the Bode plot (magnitude) of CS and cascode appended on the same plot.



Red: CS, Yellow: cascode

4) Using small signal parameters from OP simulation, perform <u>hand analysis</u> to calculate DC gain, BW, and GBW of both circuits.

DC GAIN	
CS	Cascode (with body effect on NM2)
$\boldsymbol{DC\ Gain_{CS} = A_{v_0}}$ $\boldsymbol{= -g_m(R_D \| r_o)}$	$\boldsymbol{DC\ Gain_{Cascode} = A_{v_0} = -g_{m1}\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})] \| R_D}$
$A_{v_0} = \dfrac{-g_m}{g_{ds} + \dfrac{1}{R_D}}$ $= \dfrac{-197.7\mu}{4.556\mu + 33.33\mu} = -5.21$ $20\log(5.2182) = 14.35dB$	$A_{v_0} = -g_m R_{LFD,NM2} = -g_{m1}\big(r_{o2}([g_{m2} + g_{mb2}]\, r_{o1})\big) \| R_D =$ $= -179.6\mu \times [\dfrac{[184.7\mu + 20.16\mu]}{7.053\mu \times 4.164\mu} \| 30k]$ $= -179.6\mu \times 7M \| 30k = -5.38$ $20\log(5.388) = 14.63dB$

BW	
CS	Cascode
$A_v(s) = -\dfrac{g_m(R_D \| r_o)}{1 + sR_{dom}C_{dom}}$ $\omega_{p,out} = \dfrac{1}{RC} = \dfrac{1}{(R_D \| r_o)C_L}$ $\rightarrow\ f_{p,out} = \dfrac{1}{2\pi\,(R_D \| r_o)C_L} = \dfrac{g_{ds} + 1/R_D}{2\pi C_L}$ $C_{out} = C_L + C_{db} + C_{gd} = 2.1f$ $\omega_{p,in} = \dfrac{1}{RC} = \dfrac{1}{R_{sig}C_{in}}$ $\rightarrow\ f_{p,in} = \dfrac{1}{2\pi\, R_{sig}C_{in}}$ $C_{in} = C_{gs} + A_v C_{gd} = (5.3f + 5.21 \times 0.4f) = 7.4f$ input node is dominant here	$A_v(s)$ $= \dfrac{-g_{m1}\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})] \| R_D}{1 + s\, R_{sig}\, C_{in}}$ $\omega_{p,in} = \dfrac{1}{RC} = \dfrac{1}{R_{sig}\, C_{in}} \rightarrow f_{p,in} = \dfrac{1}{2\pi R_{sig}\, C_{in}}$ input node is dominant here
$BW = \dfrac{1}{2\pi\, R_{sig}C_{in}}$ $= \dfrac{1}{2\pi(10M\Omega)(5.3f + 5.21 \times 0.4f)}$ $= 2.157MHz$	$BW = \dfrac{1}{2\pi\, R_{sig}C_{in}}$ $= \dfrac{1}{2\pi(10M\Omega)(5.3f + 5.21 \times 0.4f)}$ $= 2.157MHz$

GBW	
CS	Cascode
$GBW = \lvert A_{v_0} \rvert \times f_p$ $= g_m(R_D \| r_o) \times \dfrac{1}{2\pi\, R_{sig}C_{in}} = \dfrac{g_m(R_D \| r_o)}{2\pi\, R_{sig}C_{in}}$	$GBW = \lvert A_{v_0} \rvert \times f_p =$ $-g_{m1}\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})] \| R_D \times \dfrac{1}{2\pi\, R_{sig}C_{in}}$ $= \dfrac{g_{m1}\,[r_{o2}(1 + [g_{m2} + g_{mb2}]r_{o1})] \| R_D}{2\pi\, R_{sig}C_{in}}$
$GBW = \dfrac{g_m(R_D \| r_o)}{2\pi\, R_{sig}C_{in}} = \dfrac{5.21}{2\pi(10M\Omega)(7.4f)}$ $= 11.235MHz$	$GBW = 5.38 \times 2.157MHz = 11.61MHz$

UGF	
CS	Cascode
$let:\ \lvert A_v(s)\rvert = \frac{-g_m(R_D\|\|r_o)}{1+\boldsymbol{s}\,R_{sig}C_{in}} = 1$ $1+\boldsymbol{s}\,R_{sig}C_{in} = -g_m(R_D\|\|r_o)$ $s\,R_{sig}C_{in} = -g_m(R_D\|\|r_o) - 1$ $\boldsymbol{s} = -\omega_u = -2\pi f_u = \frac{-g_m(R_D\|\|r_o)-1}{R_{sig}C_{in}}$ $UGF = f_u = \frac{g_m(R_D\|\|r_o)+1}{2\pi\,R_{sig}C_{in}} \approx \frac{g_m(R_D\|\|r_o)}{2\pi\,R_{sig}C_{in}}$ $\approx GBW$	$1 = \frac{-g_{m1}\,[\boldsymbol{r_{o2}}(\boldsymbol{1}+[\boldsymbol{g_{m2}}+\boldsymbol{g_{mb2}}]\boldsymbol{r_{o1}})]\|\|\boldsymbol{R_D}}{1+s\,\boldsymbol{R_{sig}}\,C_{in}}$ $1+\boldsymbol{s}\,\boldsymbol{R_{sig}}\,C_{in} = -g_{m1}\boldsymbol{u}$ $\boldsymbol{s} = -\omega_u = -2\pi f_u = \frac{-g_{m1}\boldsymbol{u}-1}{\boldsymbol{R_{sig}}\,C_{in}}$ $UGF = f_u = \frac{g_{m1}\boldsymbol{u}+1}{2\pi\boldsymbol{R_{sig}}\,C_{in}} \approx \frac{g_{m1}\boldsymbol{u}}{2\pi\boldsymbol{R_{sig}}\,C_{in}}$ $\approx \frac{g_{m1}\boldsymbol{R_D}}{2\pi\boldsymbol{R_{sig}}\,C_{in}}$
$GBW = \frac{g_m(R_D\|\|r_o)}{2\pi\,R_{sig}C_{in}}$ $= \frac{5.21}{2\pi(10M\Omega)(5.3f+5.21\times 0.4f)}$ $= 11.235MHz$	$UGF = \frac{g_{m1}\boldsymbol{R_D}}{2\pi\boldsymbol{R_{sig}}\,C_{in}} = \frac{179.6\mu}{2\pi(1M)(7.4f)}$ $= 115.88MHz$

5) Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis. Comment on the results.

	Simulation		Hand Analysis	
	CS	Cascode	CS	Cascode
DC gain	32.74dB	46.83dB	32.74dB	46.1077dB
BW	725.5k	130.1k	725.11k	141.218k
GBW	31.54M	28.65M	31.465M	28.528M
UGF	31.48M	28.96M	31.465M	30.05M

Simulation and hand analysis results are close.

The input pole is the dominant **not** the output pole.

The gain bandwidth product increases for cascode amplifier.

The cascode amplifier has a little bit higher gain than CS amplifier due to having a little larger Rout.

The bandwidth of the cascode is higher than that of the CS and higher than that of cascode for gain because we reduced the miller effect by reducing cascode gain (done by using resistor RD instead of ideal current source this caused dc gain to go down but has increased circuit bandwidth).

Lab Summary

- *In Part 1 I learned*
  - How to generate and use design charts for NMOS and PMOS transistors.
  - How to design common-source and cascode amplifiers.

- *In Part 2 I learned*
  - How to do ac and DC simulations of a cascode amplifier with current-source load.
  - How the gain of a cascode amplifier with current-source load changes with frequency.
  - How to simulate the gain, the bandwidth and the GBW of a cascode amplifier with current-source load.

- *In Part 3 I learned*
  - How to do ac and DC simulations of a cascode amplifier with resistive load.
  - How the gain of a cascode amplifier with resistive load changes with frequency.
  - How to simulate the gain, the bandwidth and the GBW of a cascode amplifier with resistive load.

Acknowledgements

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[END OF LAB 3]

Kerollos Saad Thomas